Indian Institute Of Technology Madras
Department of Mechanical Engineering

KINEMATICS AND DYNAMICS OF MACHINERY

COURSE PROJECT

Submitted by:

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ME21B005

Submitted to:

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PROJECT OVERVIEW

- To introduce the mechanism used, it is a basic four-bar mechanism. The bottom of the box, link CD, is the coupler in the aforementioned four-bar mechanism.
- As mentioned in the problem statement, I have assumed an appropriate scale in the units of the CAD softwares used.

  1 cm = 1 unit
- The initial iteration of the design for the mechanism was done on AUTOCAD. I made the scaled workspace, and began with the initial ideation.
- The initial ideation was of two-position synthesis considering the initial and final positions of points C and D; finding their perpendicular bisectors and locating some suitable center on these bisectors inside the shaded region for the fixed pivots.
- The selected pivot points are marked as O2 and O4.
- O4 is joined to O2; and is link number 1.
- O2 is joined to C; and is link number 2.
- C is joined to D; and is link number 3.
- D is joined to O4; and is link number 4.
- Therefore, it is a four-bar mechanism, with link 1 grounded, and link 3 as the coupler.

MECHANISM ANALYSIS

- **Grashof Analysis**
  - Longest link length = 194.48cm = L
  - Shortest link length = 49.71cm = S
  - Intermiediate link lengths: 163.91 = P, 91cm = Q

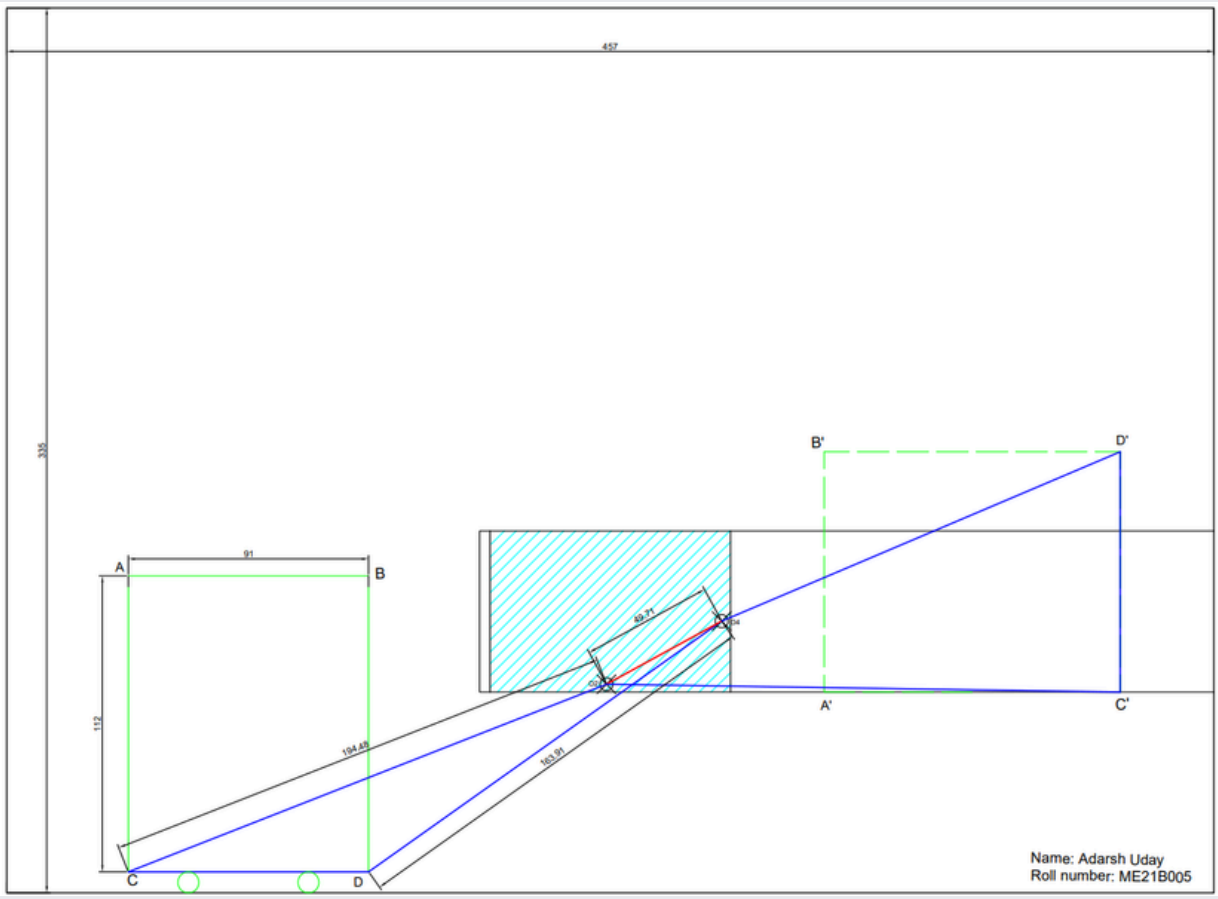
  - Clearly, S + L < P + Q
  - Therefore, this mechanism follows Grashof's law, and is thus a **Grashof mechanism**.

  - Since the shortest link is grounded, this Grashof mechanism is a *Double Crank* mechanism.
  - Link 1 is grounded.
  - Links 2 and 4 are both "Cranks".
  - Link 3, the coupler, "tumbles" due to the motion of the two cranks.
  - Here is a video showing the motion of a generalised Double Crank mechanism. It is quite evident that the coupler rotates 360° about its own center of rotation.

- **Tranmission Angles**
  - The transmission angle is the angle between the coupler and the output link.
  - Transmission angle at initial extreme position is 145°
  - Transmission angle at final extreme position is 67°

IDEATION AND CONSTRUCTION

- I began with the initial idea of two-position synthesis considering the initial and final positions of points C and D on the box.
- Then, I joined the initial and final points of C (C and C' respectively), and also joined the initial and final points of D (D and D' respectively).
- I constructed perpendicular bisectors to each of the lines CC' and DD', and then considered one point on each bisector that lies in the fixed region.
- The point on the bisector of CC' is O2 and the point on the bisector of DD' is O4.
- O2 and O4 are now the fixed pivots for the four-bar mechanism.



AUTOCAD construction diagram

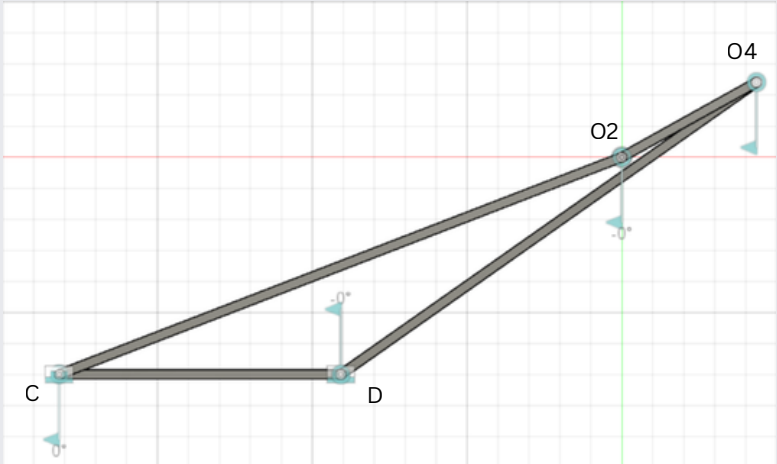
DELIVERABLES

- The motion of the proposed four-bar mechanism does not interfere with the smooth operation of the truck.
- The motion of the container is smooth as the mechanism operates from extreme to extreme.
- The proposed mechanism is formulated with all given constraints taken into consideration, and is entirely contained inside the given workspace.
- The links are all rigid, the joints are all of single degree of freedom (revolute pin joints)
- Recommendations for large-scale production:
  - Using a better connecter between couplers of both mechanisms that sits the box on itself well, and releases the box at the end.
  - Possibly using washers instead of screws, such that the ends of these joints that stick out do not interfere with the other links, and such that these joints allow fully free rotation of cranks & coupler.
  - Fixing the two ground positions slightly better. I feel the ground position I have chosen is tedious to demonstrate, and is also probably not feasible for large scale produce.

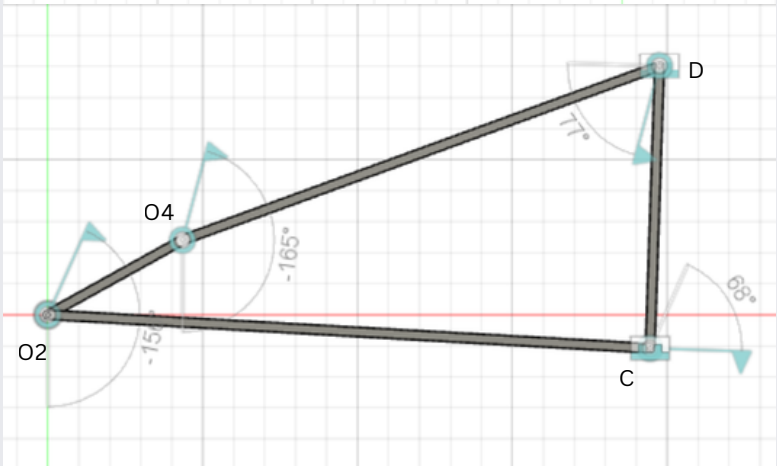
MODEL

Model Animation

Initial position



Final position



Physical Model

Initial position
The box sits on the a support fixed between both couplers. Therefore, the base of the box, which is on this support, performs the exact same motion as the couplers themself do.
The base of the box is horizontal in the initial position.

Final position
Crudely speaking, the box performs a 270° rotation. Both of the couplers are vertical in the final position.
The base of the box is also vertical, hence satisfying the problem statement.

Videos of the working model

I have attached all model videos, designs and animations of the proposed mechanism in this Google Drive link.
I have also submitted the videos of the physical model, PDF of the proposed design, animation of the design and this report directly to Google Classroom. The link is open to any IITM email ID.

Drive Link :

https://drive.google.com/drive/u/1/folders/1EXkkG-qxQuxO_YuUHnMDE21oDi_NtH11